ATML COURSE PROJECT · PROJECT DESCRIPTION

Seeing Through the Clutter:

Disentangling Distractors in Vision-Language Models

Group 16 · Advanced Topics in Machine Learning · November 2026
Zoya Khan · Zehra Talat (27100407) · Maryam Zafar (27110054) · Mohid Ali (28100043)

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1 Summary

Vision-language models (VLMs) answer questions about clean images well, but often fail when the same image also contains clutter: extra objects, look-alikes or misleading text. This project asks why, and whether it can be fixed without retraining the model.

Our hypothesis is that the model's internal image representation mixes (entangles) clutter information with the information needed for the answer. We will measure how much clutter hurts, test whether the representation is entangled, and then use a sparse autoencoder to separate the two and switch off the clutter.

2 Problem statement

Real images rarely contain only what a question is about. A model asked to count apples must ignore a tomato next to them, the printed words "3 apples", and unrelated objects. Current VLMs are easily misled by such clutter, but it is unclear whether the failure comes from what the model sees or from how it stores what it sees.

Studying this needs controlled data. In natural photos, adding clutter also changes the rest of the scene, so a drop in accuracy cannot be blamed on the clutter alone. We therefore build images in which only the clutter changes and the correct answer stays exactly the same.

3 Hypothesis and research questions

Hypothesis. When an image mixes task-relevant content with clutter, the VLM's image representation entangles the two, and this entanglement causes wrong answers. Separating the representation and removing the clutter part should raise accuracy.

- **RQ1, effect:** How much does each kind of clutter lower accuracy, and which kinds hurt most?
- **RQ2, entanglement:** Does clutter information leak into the representation of the target objects even when their pixels are unchanged, and does that leak cause errors?
- **RQ3, disentanglement:** Can a sparse autoencoder separate task-relevant features from clutter features, so that switching off the clutter features improves accuracy?

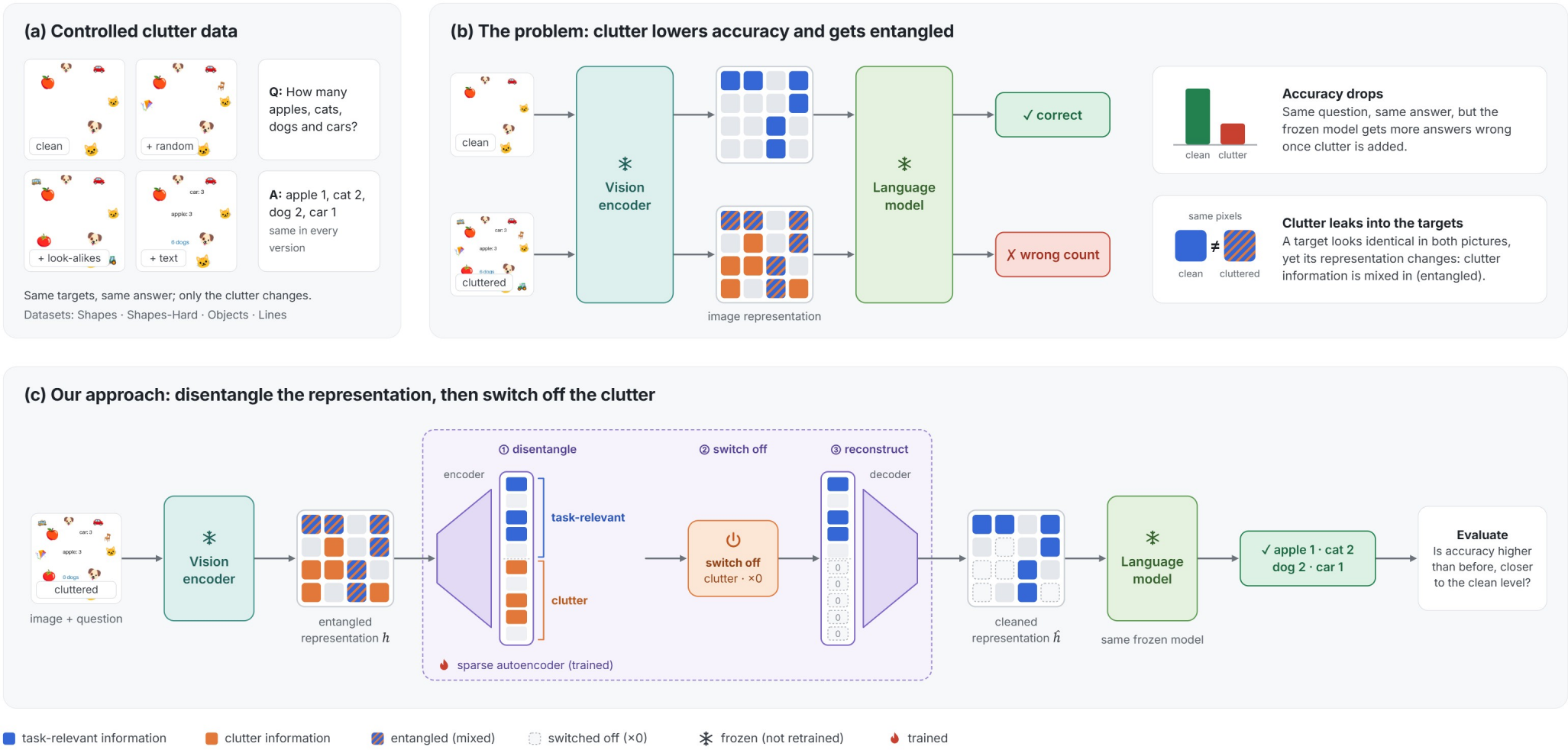


Figure 1. Overview of the project. (a) Each scene is drawn clean and with different clutter (random objects, look-alikes, misleading text); the targets, the question and the answer stay identical, so any change in the model's behaviour is caused by the clutter. (b) We measure how much clutter lowers a frozen vision-language model's accuracy, and test whether clutter information leaks into the representation of the targets even though their pixels are unchanged (entanglement). (c) We then split the entangled image representation into separate features with a sparse autoencoder, switch off the features that carry clutter, rebuild the representation, and let the same frozen model answer from it. The goal is higher accuracy than without the fix, moving toward the clean level (it does not have to match it).

4 Datasets

We built four synthetic datasets (12,500 images) in which each scene is drawn once clean and several times with different clutter. The targets are pixel-identical across versions, so the correct answer never changes and any accuracy drop is caused by the clutter.

Table 1. The four controlled clutter datasets.

Dataset	Task	Clutter versions	Images
Shapes-VQA	Count circles, triangles, squares and stars	squiggles, blobs, look-alikes (broken or wavy shapes)	2,000 500 scenes × 4
Shapes-VQA Hard	Count only the shapes of one named colour	wrong colour, look-alikes, misleading text, squiggles of rising complexity, other named shapes, all combined	5,000 500 × 10
Objects-VQA	Count apples, cats, dogs and cars	random objects, look-alikes (tomato, tiger, wolf, bus…), misleading text, all combined	2,500 500 × 5
Lines-VQA	Follow a line from a dot to its end box	other lines with 0, 2, 4 or 8 crossings; coloured distractor lines	3,000 500 × 6

Splits are by scene (350 / 75 / 75 scenes for train / validation / test). Every label is checked automatically against the image pixels.

5 Method

The method has three stages, one per research question, shown in the project block diagram (Figure 1). The VLM itself is never retrained; only a small sparse autoencoder is trained.

1. **Measure the effect (RQ1).** Run frozen VLMs on every version and compare accuracy with the clean version of the same scenes. The main model is Qwen3-VL-4B; InternVL3.5, Molmo2 and Gemma 4 check that the findings hold across models.
2. **Show the entanglement (RQ2).** Compare the model's representation of target regions in the clean and cluttered image. Their pixels are identical, so any difference is clutter that leaked in. A simple probe tests whether clutter can be read back from the targets, and swapping in the clean representation tests whether the leak causes wrong answers.
3. **Disentangle and suppress (RQ3).** Train a sparse autoencoder that splits the image representation into many separate features. Using the known object locations, find the features that respond to clutter, set them to zero, rebuild the representation and let the same frozen model answer.

6 Evaluation and success criteria

Success means accuracy on cluttered images is clearly higher after switching off the clutter features than before; it does not have to reach the clean level. The main measure is the share of the lost accuracy that comes back:

$$\textit{gap closed} = \frac{\mathrm{acc}_{\mathrm{after}} - \mathrm{acc}_{\mathrm{cluttered}}}{\mathrm{acc}_{\mathrm{clean}} - \mathrm{acc}_{\mathrm{cluttered}}}$$

- **Controls:** switching off the same number of random features should not help, and accuracy on clean images should not drop.
- **Statistics:** 95% bootstrap intervals over scenes and paired McNemar tests, as in our earlier study.
- **Predictions, fixed before running:** the largest gains for misleading text and unrelated objects, smaller gains for look-alikes, which share features with the real targets.

7 Preliminary findings

Our first study confirms that the accuracy drop is real. In all four datasets and in every model we tested, adding clutter lowered accuracy, often sharply, even though the correct answer never changed.

Clutter that resembles the targets hurts most. On Objects-VQA, for example, the main model answers almost every clean image correctly, but fewer than a third once look-alikes are added. Models also copy misleading text into their answers.

Two findings shape the next stages. Models mostly guess instead of tracing lines, so Lines-VQA cannot test entanglement. Objects-VQA, where clean accuracy is highest, becomes the main test set.

8 Expected contributions and risks

Contributions

- Four controlled clutter datasets with pixel-checked labels that others can reuse.
- A direct test of whether VLMs entangle clutter with task information, using targets whose pixels never change.
- A fix that needs no retraining of the VLM: removing clutter features with a sparse autoencoder.

Risks and fallbacks

- **The information may be lost rather than mixed.** If a probe cannot tell a look-alike from a real target, the limit is perception; we would report that as the finding.
- **The leak may be harmless.** If errors come from the clutter regions themselves, we switch off those regions instead of untangling the target representation.
- **Look-alikes may share features with real targets.** Switching them off could also remove real objects; we will measure and report this trade-off.

References

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